CS 1101 Unit 5

Programming Assignment

Assignment Response

1. **String Manipulation and Iteration**

Write a program to display your name and perform the following operations on it:

- Display n characters from left. (Accept n as input from the user)
- Count the number of vowels. Reverse it.

```
    # String Manipulation and Iteration

def main():
    # 1. Display name and take input for 'n' characters
    myName = "Sona Hakobyan"
    print(f"Original Name: {myName}")

    try:
        n = int(input("Enter the number of characters to display from the left: "))
        # Slicing from index 0 to n
        leftChars = myName[:n]
        print(f"The first {n} characters are: '{leftChars}'")
    except ValueError:
        print("Invalid input. Please enter an integer.")

    # 2. Count the number of vowels using iteration
    vowels = "aeiouAEIOU"
    vowelCount = 0
```

```python
    for char in myName:
        if char in vowels:
            vowelCount += 1
    print(f"Number of vowels in the name: {vowelCount}")

    # 3. Reverse the name using string slicing
    reversedName = myName[::-1]
    print(f"Name in reverse: {reversedName}")

# This correctly calls the function without causing an infinite loop
if __name__ == "__main__":
    main()
```

```
~/Doc/UoPeopleCS1101-01ProgrammingFundamentals/Programming Assignment Unit 5 | main +1 ?1    python3 nameStringOperations.py
Original Name: Sona Hakobyan
Enter the number of characters to display from the left: 4
The first 4 characters are: 'Sona'
Number of vowels in the name: 5
Name in reverse: naybokaH anoS
```

Technical Explanation

This program demonstrates fundamental string operations and iteration in Python. A string is defined as a *"sequence, which means it is an ordered collection of other values"* (Downey, 2015, p. 71). Each character is an item identified by an integer index, which in Python always starts from 0.

First, the program extracts a specific number of characters from the left based on user input. *A segment of a string* like this *is called a slice* (Downey, 2015, p. 73). By evaluating myName[:n], the program selects a slice from index 0 up to the nth index.

Next, the program counts the number of vowels in the name. This is achieved through iteration, which is the *"repeated execution of a set of statements"* (Downey, 2015, p. 69). The for loop performs a traversal, meaning it processes the string one character at a time from beginning to end. To track the vowels, a counter variable (vowelCount) is initialized to zero. During the traversal, the in boolean operator checks if the current character appears as a substring in the vowels string. If it does, the counter undergoes an increment, updating its value by adding one.

Finally, the program reverses the name using a slice with a negative step [::-1]. It is crucial to understand that strings are immutable, meaning we cannot change the items within an existing string object. Therefore, the slicing operation does not modify myName, but instead returns a completely new string sequence.

242 words

References

Downey, A. (2015). Think Python: How to think like a computer scientist. Green Tree Press. https://greenteapress.com/thinkpython2/thinkpython2.pdf